

Territory-Aware Discrete-Time Deep Learning for Ischemic Stroke Recurrence and Mortality Risk Prediction with Patients Risk Trajectories Using Longitudinal EHR Data

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Motivation

Ischemic stroke is a leading cause of death and long-term disability worldwide [1]. Accurate post-stroke risk prediction is critical for guiding secondary prevention and personalized care.

10–20% of stroke patients experience another stroke within 2 years [2]
>15% die within 2 years of their initial event [2]

Gaps in Current Approaches

1 Binary endpoints

Fixed timepoint classification obscures temporal risk dynamics [3]

2 Pooled vascular territories

Anterior vs. posterior recurrence patterns are not modeled [4-5]

3 Cox/PH assumptions break down

Real-world cohorts show nonlinear, time-dependent effects that violate proportional-hazards assumptions [6,7].

Contributions

Territory-Aware Discrete-Time Framework

Models' recurrence and mortality as interval-specific hazards across three clinically meaningful phases (0–90, 91–365, 366–730 days), explicitly incorporating vascular territory.

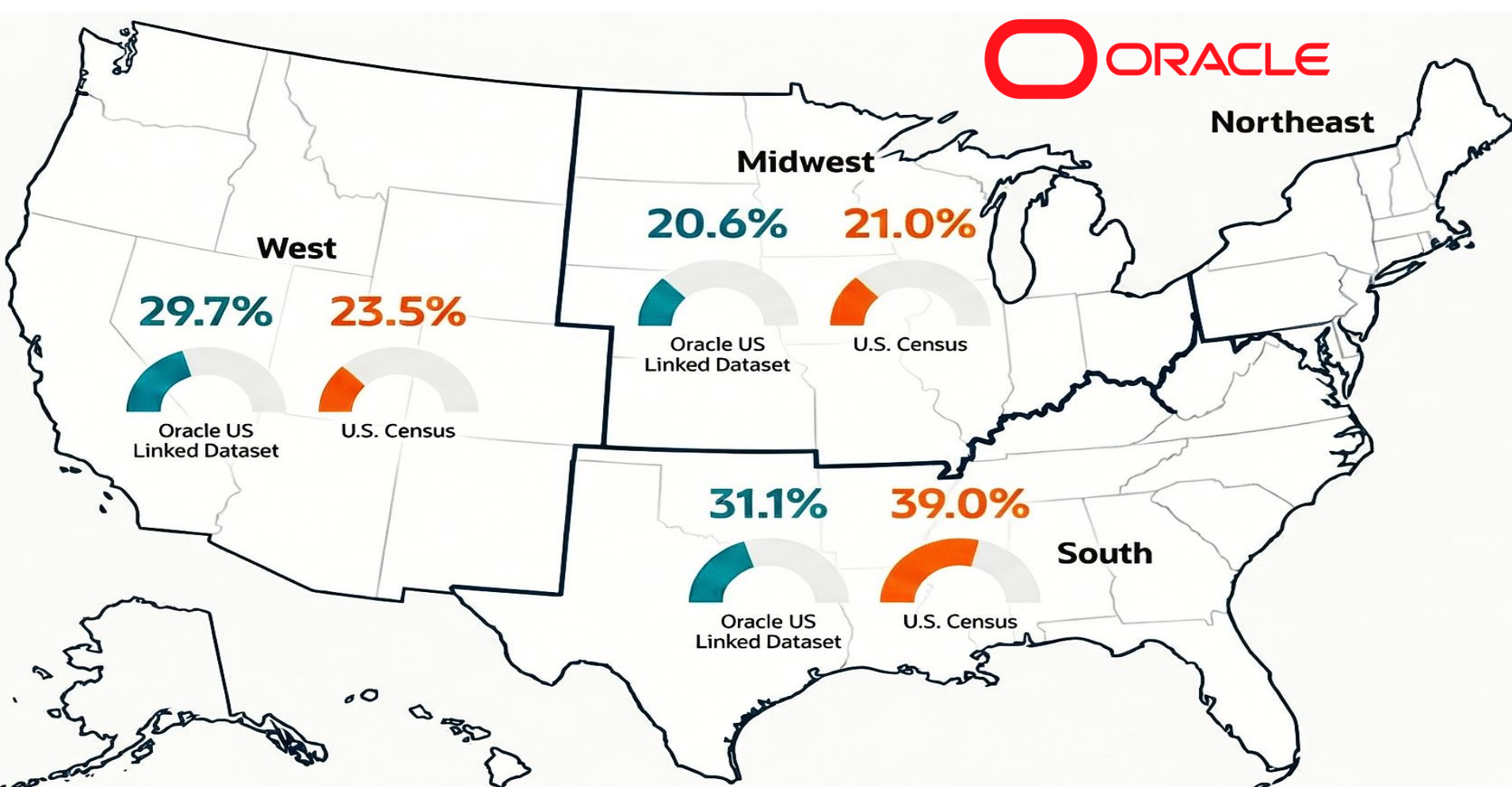
Time-Resolved Risk Stratification

Translates continuous model predictions into clinically interpretable risk categories (Low, Moderate, High, Very High) and patient-level trajectories.

Large-Scale Real-World Evaluation

118,727 ischemic stroke patients across 149 U.S. health systems [8]. Evaluated discrimination, calibration, O/E agreement, and individual-level trajectories.

Data Descriptions



Vascular Territory

ACIS · Anterior Circulation

Vascular source: Internal carotid arteries

Cohort: n = 88,697 (74.7%)

Key clinical pattern

Higher mortality (14.4%) and same-territory recurrence (15.3%)

PCIS · Posterior Circulation

Vascular source: Vertebrobasilar system

Cohort: n = 30,030 (25.3%)

Key clinical pattern

~3× higher different-territory recurrence (1.7% vs 0.5%)

TA-DT-DL Framework

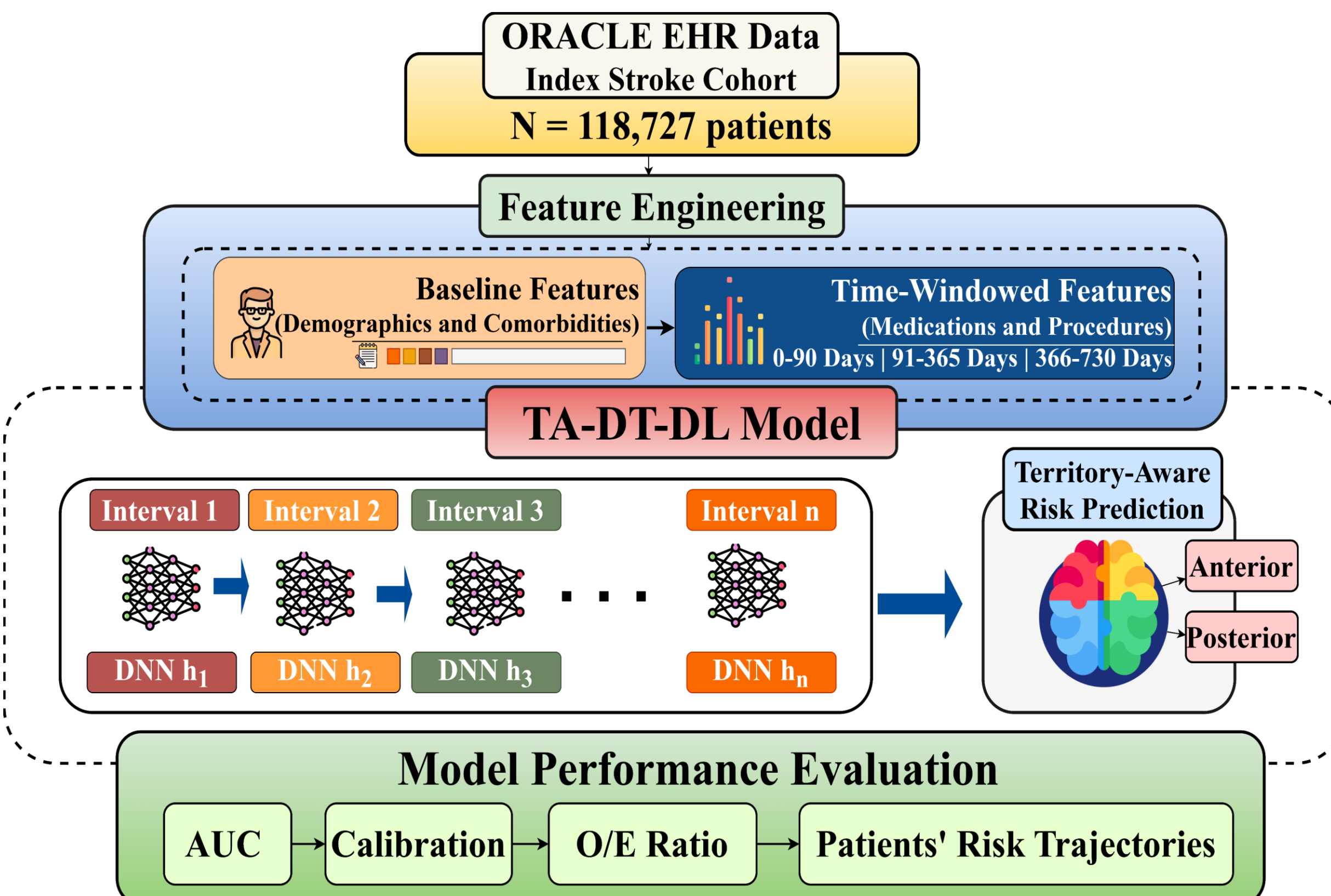
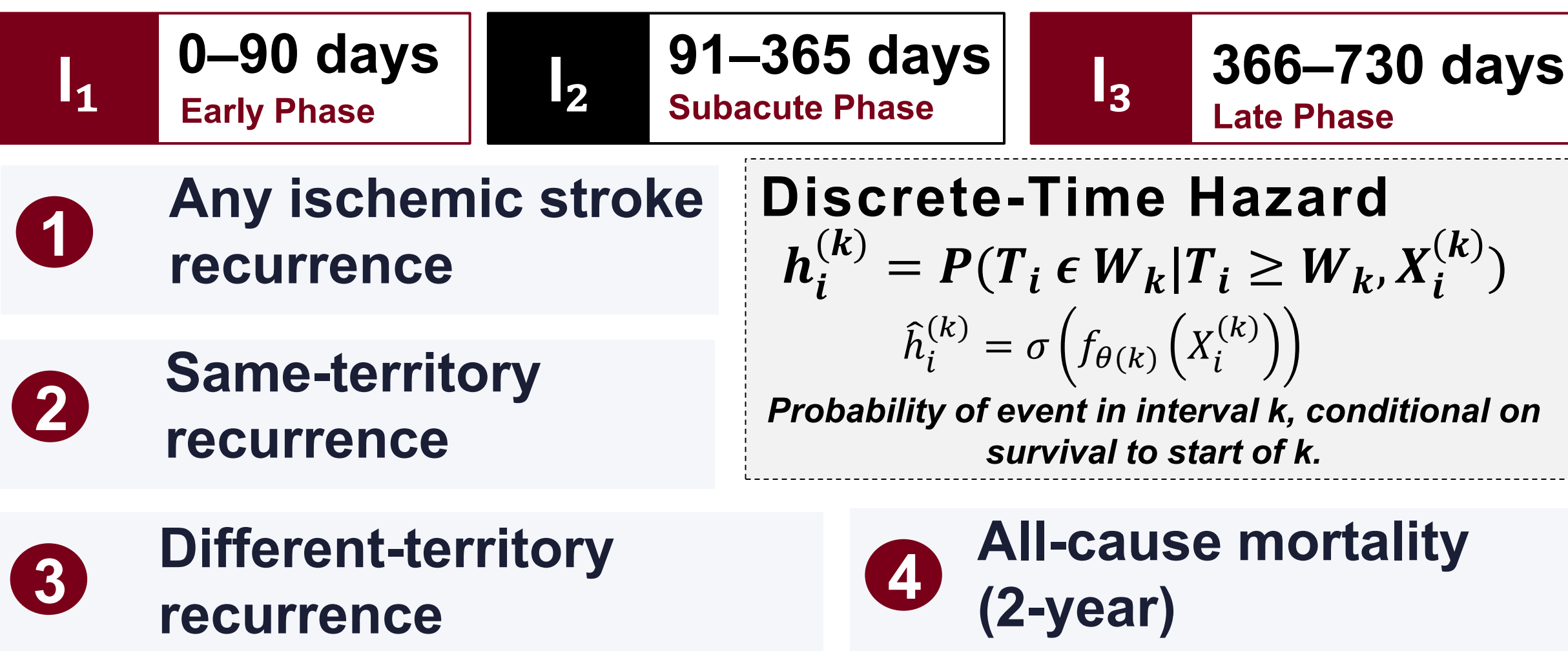


Fig. 1. Proposed architecture of the TA-DT-DL framework.

Methods

Discrete-Time Intervals



Observed Outcome

Table 1. Territory-Stratified 2-Year Observed Outcomes (95% CI)

Outcome	Anterior (n = 88,697)	Posterior (n = 30,030)
Any Recurrence	15.7% (15.5–16.0)	15.1% (14.7–15.6)
Same-Territory	15.3% (15.1–15.6)	12.9% (12.5–13.3)
Different-Territory	0.5% (0.5–0.6)	1.7% (1.5–1.8)
Mortality	14.4% (14.1–14.6)	11.2% (10.9–11.6)
Survived	85.6% (85.4–85.9)	88.8% (88.4–89.1)

Results

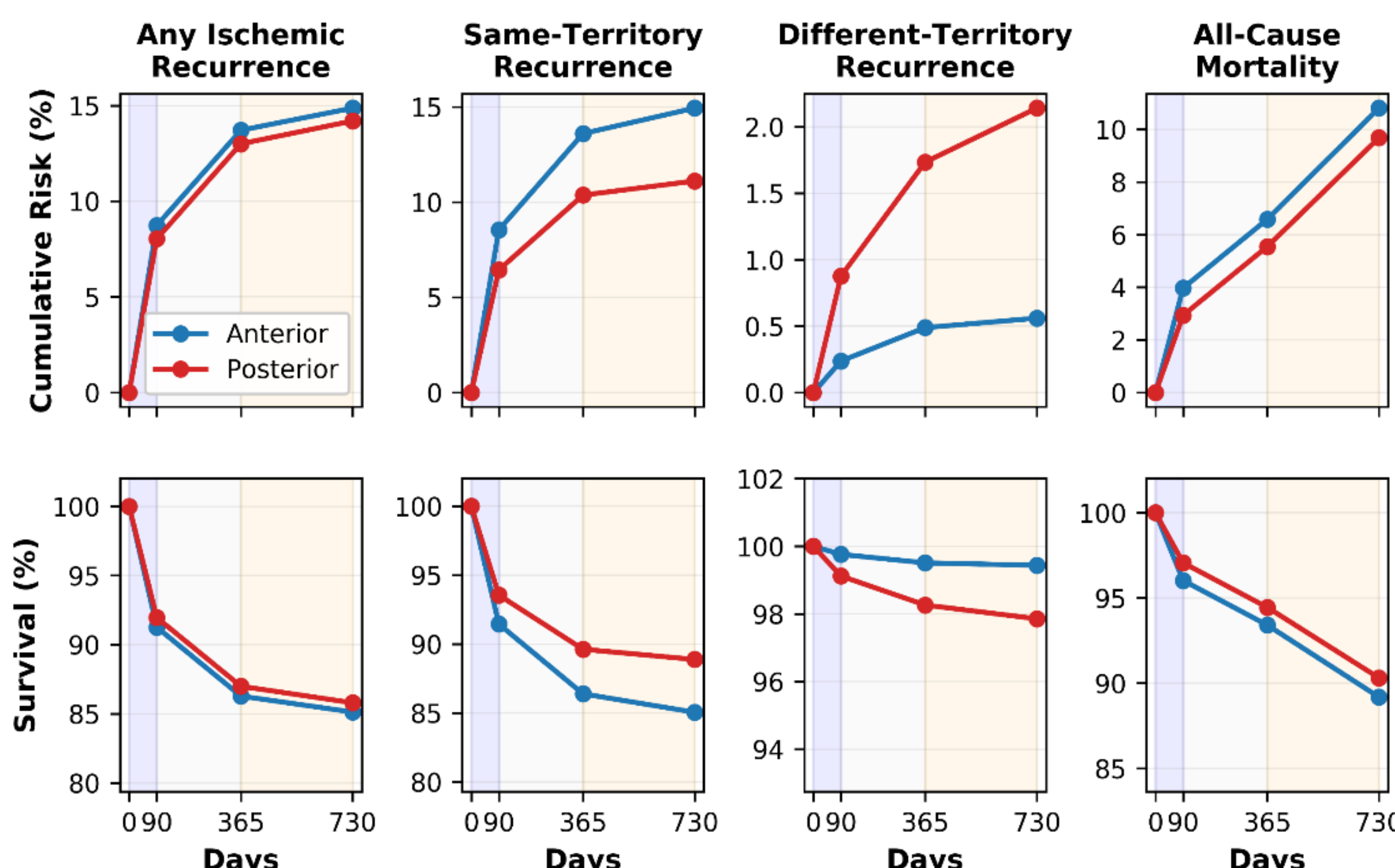


Fig. 2. Territory-specific interval-wise predicted risk and survival trajectories.

ACIS	PCIS	BOTH
Steeper rise in same-territory recurrence and mortality	Markedly higher different-territory recurrence (~3×)	Sharpest rise occurs in the 91–365-day subacute window

Subacute window (91–365 days) shows the steepest rise in recurrence and mortality across both territories.

Calibration

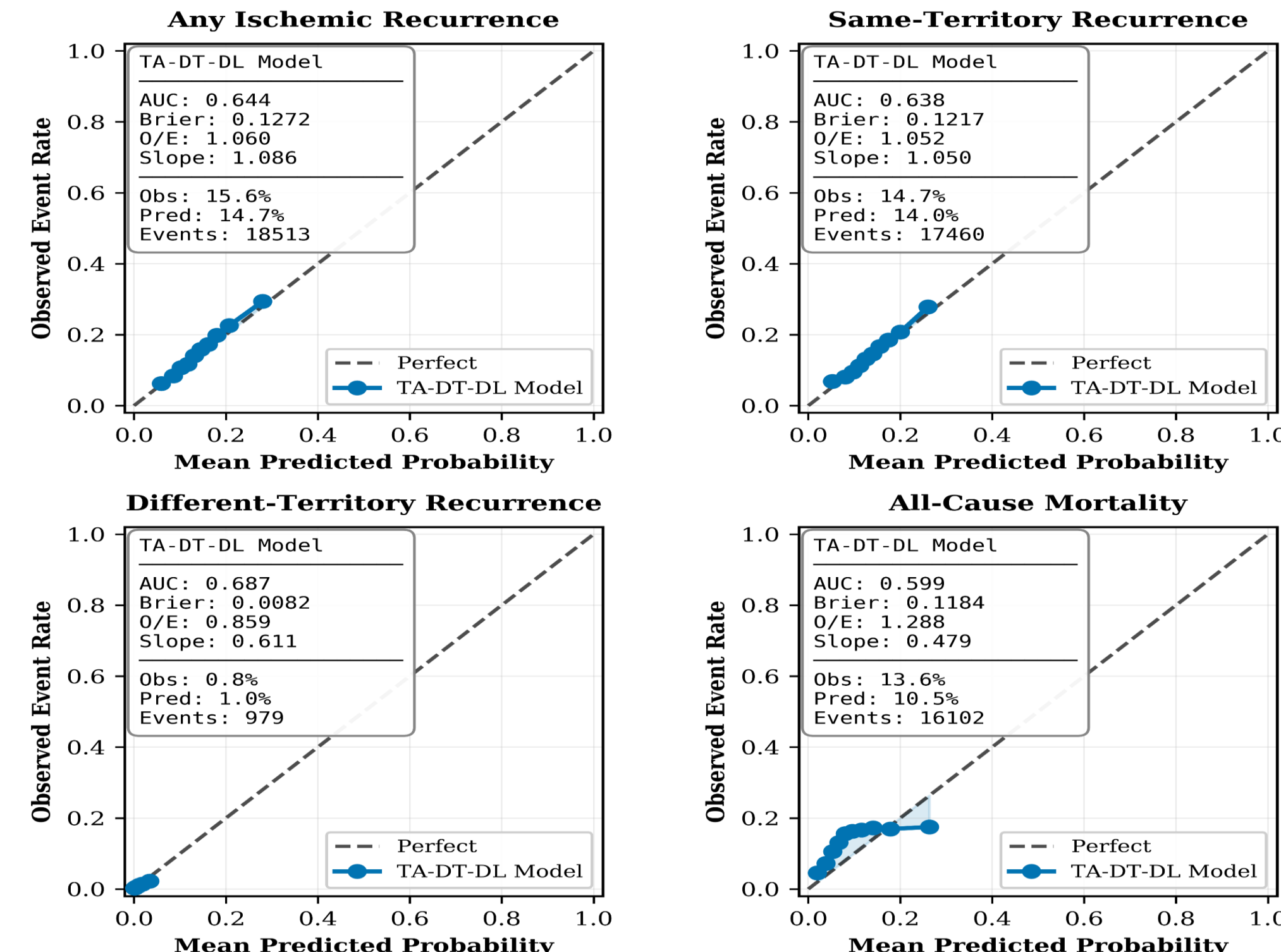


Fig. 3. Calibration of TA-DT-DL model for stroke recurrence and mortality across four outcomes.

Performance

Table 2. Performance comparison of TA-DT-DL with baseline, tabular, and deep models.

Performance Metric	TA-DT-DL	RNN	LSTM	Cox/LR	XGBoost
MAE (percentage points)	1.2	1.4	1.44	8.99	38.04
Deviation from perfect calibration (O/E = 1.0)	0.132	0.241	0.212	0.419	0.786
Mean AUC	0.642	0.730	0.651	0.743	0.763
Well-calibrated outcomes (O/E 0.90–1.10)	4/8 (50%)	1/8 (12.5%)	2/8 (25%)	2/8 (25%)	0/8 (0%)
Acceptable outcomes (O/E 0.80–1.20)	6/8 (75%)	5/8 (62.5%)	5/8 (62.5%)	3/8 (37.5%)	0/8 (0%)
Best performing outcomes	7/8 (87.5%)	1/8 (12.5%)	0/8 (0%)	2/8 (25%)	0/8 (0%)

KEY INSIGHT

TA-DT-DL achieves the lowest calibration error (1.2 pp) and the most stable O/E range, discrimination is not enough for clinical deployment.

Patient-Level Trajectories

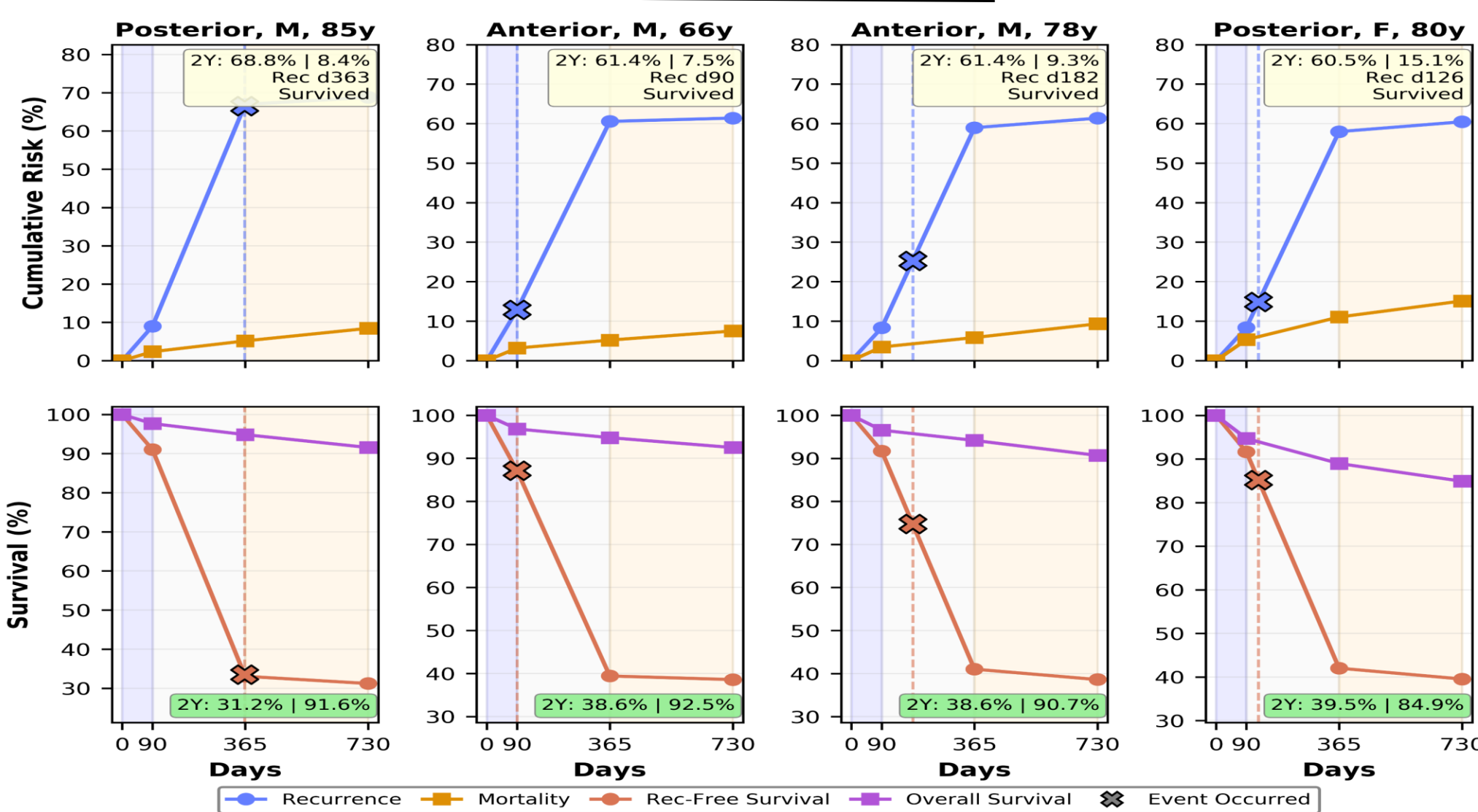


Fig. 4. Individual high-recurrence risk trajectories with illustrative patient-level cases.

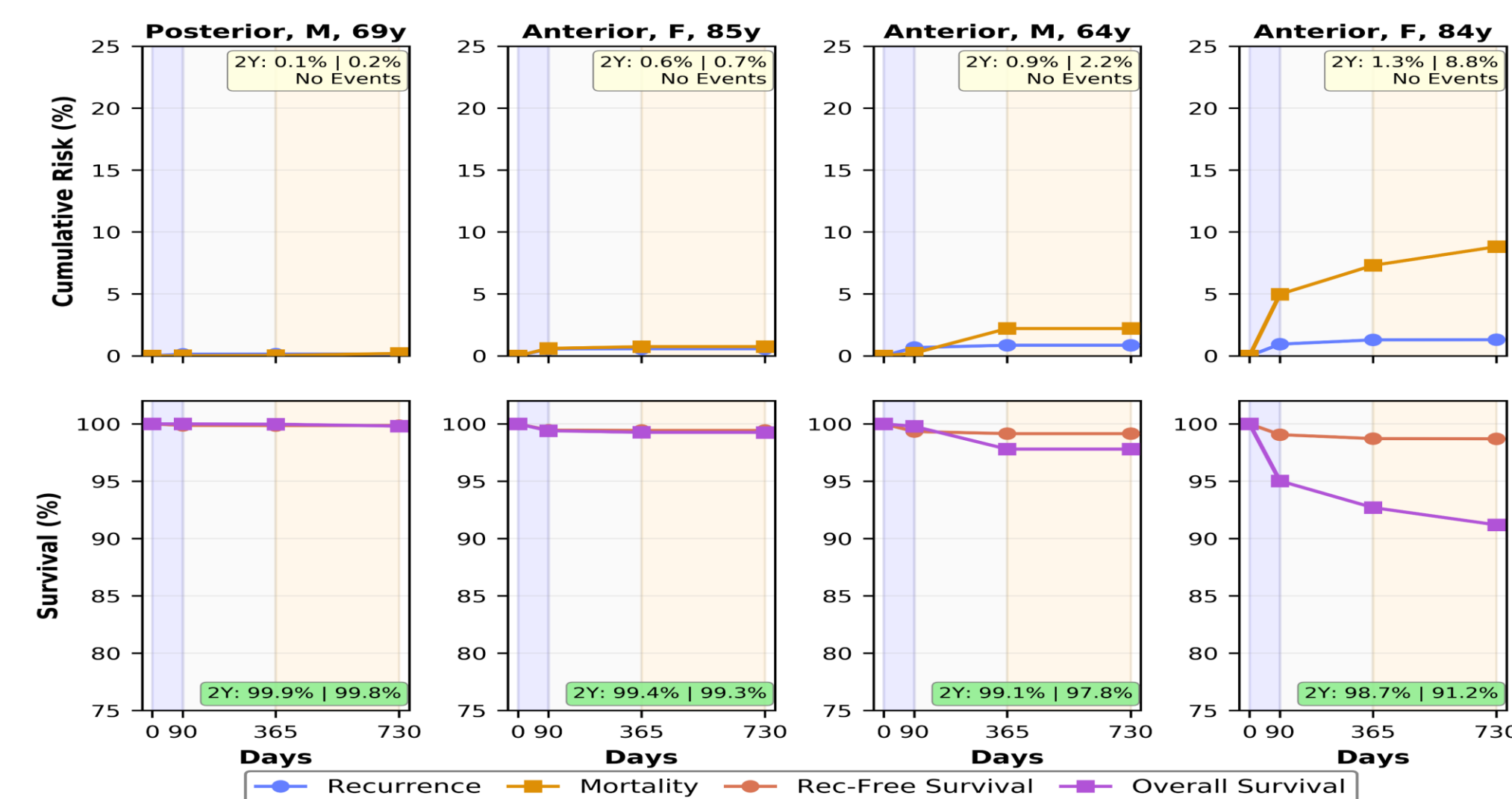


Fig. 5. Individual low-recurrence risk trajectories with illustrative patient-level cases.

Clinical Significance

- ✓ Personalized Follow-Up
- ✓ Subacute-Phase Vigilance
- ✓ Territory Specific Care
- ✓ Reliable Probabilities

Conclusion

Integrates vascular territory with interval-specific hazards over 2 years. Most reliable calibration (1.2 pp error), critical for clinical deployment. Identifies 91–365-day subacute window as critical recurrence escalation period.

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